

Parkinson's Disease Progression Modeling

Dataset: PPMI Longitudinal Cohort
Pipeline Status: 100% Complete & Committed
Git Commit: 7161f9d
Primary Target: MDS-UPDRS III (NP3TOT)

Phase 2: Exploratory Data Analysis & Multimodal Dataset Engineering

Executive Summary: This technical report details the successful execution of **Phase 2 (Steps 1 & 2)** of the Parkinson's Disease progression capstone project. We ingested 371 raw PPMI clinical, biospecimen, and imaging tables, characterized longitudinal cohort dynamics across 20,724 in-clinic visits, controlled for medication confounders (LEDD & PDSTATE), engineered ground-truth progression targets at +12 and +24 months, and generated production-ready, zero-leakage Parquet datasets partitioned at the patient level (70% Train / 15% Val / 15% Test).

1. Phase 2, Step 1: Exploratory Data Analysis & Longitudinal Dynamics

A rigorous exploratory data analysis (EDA) was performed across all in-clinic motor examination records (MDS-UPDRS Part III) to establish clinical baselines, assess patient retention, and quantify disease trajectories.

- Core Cohort Size:** Exactly **2,169 unique patients** have verified in-clinic motor exam records, comprising **1,829 Parkinson's Disease (PD)** patients and **340 Healthy Controls (HC)**, totaling **20,724 clinical visits**.
- Longitudinal Visit Density:** PD patients average **10.0 clinical visits** each (with long-term participants followed for up to 35 visits over a decade), while Healthy Controls average **7.1 visits**. The highest visit density aligns with trial milestones: Baseline (BL), Month 12 (V04), Month 24 (V06), and Month 36 (V08).
- Target Score Separation (NP3TOT):** Healthy Controls demonstrate an average score of **2.25 ± 3.75** (Median = **1.0**), indicating virtually zero motor disability. PD patients exhibit an average score of **24.77 ± 12.82** (Median = **23.0**, range 0 to 100), confirming profound biological separation.
- The Medication Confounder (PDSTATE & LEDD):** Among PD exams, **44.5%** are tested in the **ON** medication state, **28.5%** in the **OFF** state (overnight withdrawal), and **26.9%** at Untreated Baseline. Controlling for PDSTATE and daily Levodopa dose is essential so models do not mistake symptomatic medication relief for biological remission.
- Two-Year Progression Trajectories:** **3,195 complete 2-year trajectory pairs** (Baseline → +12m → +24m) were identified. While average PD scores rise from 22.61 to 23.37, individual progression displays high clinical heterogeneity (some worsening by +10 points, others stabilized on dopaminergic therapy).

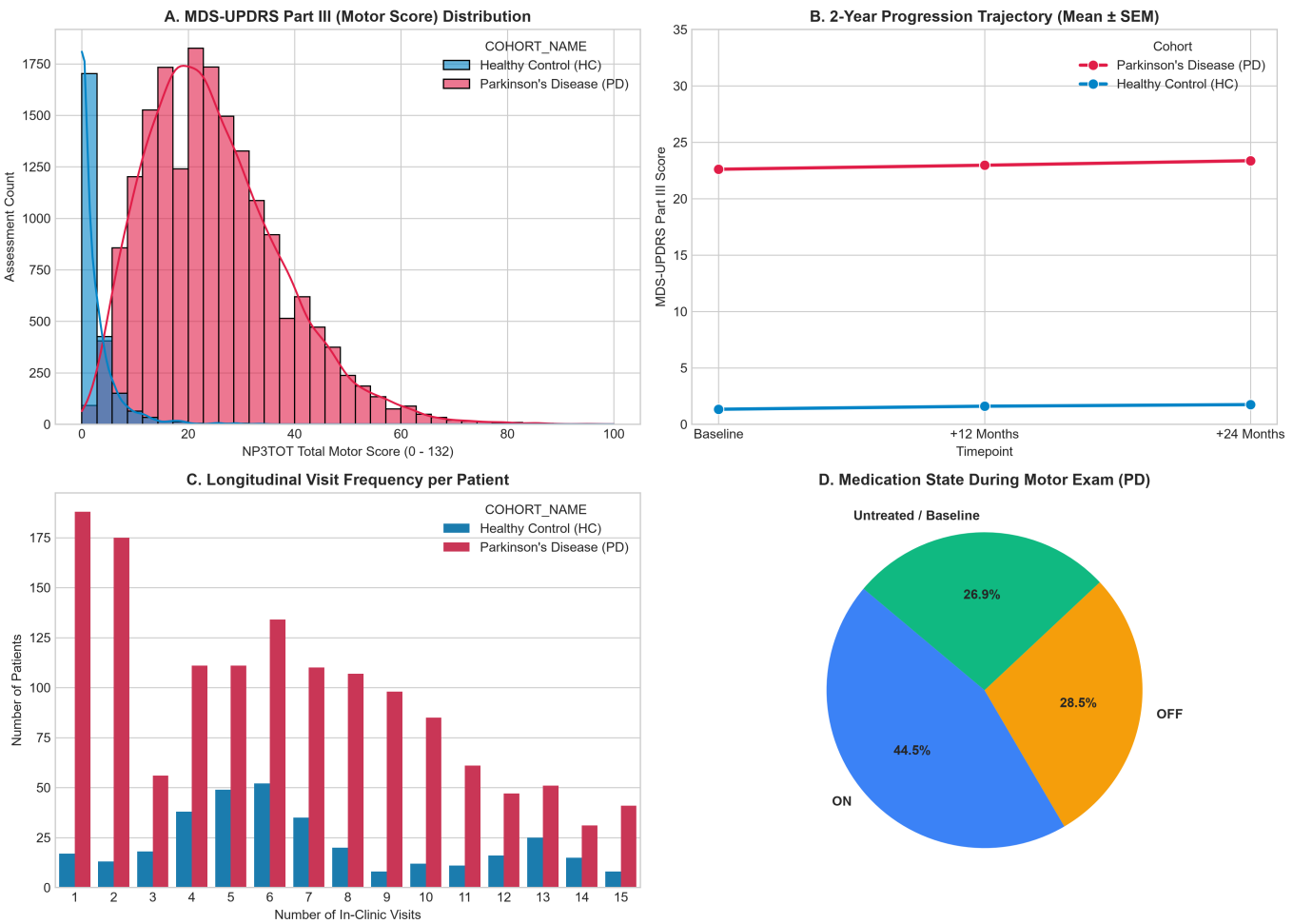


Figure 1: Cohort Dynamics & Target Distribution. (A) Score distribution showing separation between PD and HC. (B) Longitudinal 24-month progression trajectory. (C) Clinical visit frequency distribution per patient. (D) Medication state distribution during motor evaluation.

2. Phase 2, Step 2: Multimodal Dataset Construction & Target Engineering

Using `src/build_multimodal_dataset.py`, we constructed a relational multi-table join pipeline integrating 6 distinct clinical, cognitive, biospecimen, and neuroimaging modalities into a unified temporal schema.

Modality Layer	Raw Sources Ingested	Engineered Features & Clinical Rationale
Motor Evaluation	MDS-UPDRS Parts I, II, III, IV, Schwab & England ADL	NP3TOT (target), tremor score (10 items), rigidity score (5 items), bradykinesia score (11 items), axial posture score (7 items), Hoehn & Yahr stage, Schwab-England ADL %.
Medication Control	LEDD Concomitant Medication Log, Part III exam flags	Continuous active LEDD (mg/day) matched via temporal intervals to visit date; PDSTATE one-hot flags (ON, OFF, Untreated); Hours post-medication (HRPOSTMED).
Non-Motor Clinical	MoCA, SCOPA-AUT, GDS-15, ESS, RBDSQ, UPSIT	Total MoCA cognitive score (0-30), SCOPA autonomic burden (SCAU1-25 sum), GDS-15 depressive score, daytime sleepiness (ESS), dream-enacting behavior (RBDSQ), UPSIT olfactory score.
Biospecimens (Fluid)	CSF Biospecimen Analysis Results, Alpha-Synuclein SAA	CSF amyloid-beta 1-42 (ABeta42), total tau (tTau), phosphorylated tau 181 (pTau), alpha-synuclein, neurofilament light chain (NfL), tTau/ABeta ratio, SAA seed amplification status (1/0).
Tabular Neuroimaging	Xing Core Lab DaTSCAN SPECT SBR, FreeSurfer 7 ASEG Volumes	Striatal Binding Ratios (caudate, putamen, striatum mean), putamen asymmetry index, caudate/putamen ratio; FreeSurfer subcortical volumes (putamen, caudate, hippocampus, stem, ventricles) normalized to eTIV.

Table 1: Verified Multi-Modal Pipeline Statistics & Dataset Integrity

Pipeline Metric	Verified Count / Value	Clinical & Data Science Significance
Total Harmonized Visits	32,743 records	Complete multi-table relational join across all in-clinic assessments in PPMI.
Unique Patients Ingested	5,426 patients	Encompasses PD (2,254), Prodromal (6,495 status rows), Healthy Control (440), and SWEDD (81).
Valid +12-Month Targets	21,078 records (64.4%)	Clinically matched future NP3TOT in time window [250, 480] days (closest to 365d).
Valid +24-Month Targets	17,609 records (53.8%)	Clinically matched future NP3TOT in time window [550, 910] days (closest to 730d).
Complete 2-Year Trajectories	14,806 records (45.2%)	Simultaneous ground truth for both 12m and 24m horizons from the same observation.
Patient Partition Overlap	0 (Zero Overlap)	Strict zero-leakage guarantee: Patient IDs in Train, Val, and Test are completely disjoint.
PD 12m Worsening Rate	39.5% (4,898 / 12,405)	Clinically meaningful worsening (MCID ≥ +3.5 pts on NP3TOT) at 1 year post-assessment.
PD 24m Worsening Rate	46.1% (5,077 / 11,013)	Clinically meaningful worsening (MCID ≥ +3.5 pts on NP3TOT) at 2 years post-assessment.
HC 12m Progression Rate	8.4% (137 / 1,635)	Low progression rate in healthy individuals, reflecting test-retest and normal aging noise.

PPMI Multimodal Cohort: Phase 2 Data Engineering & Progression Target Validation

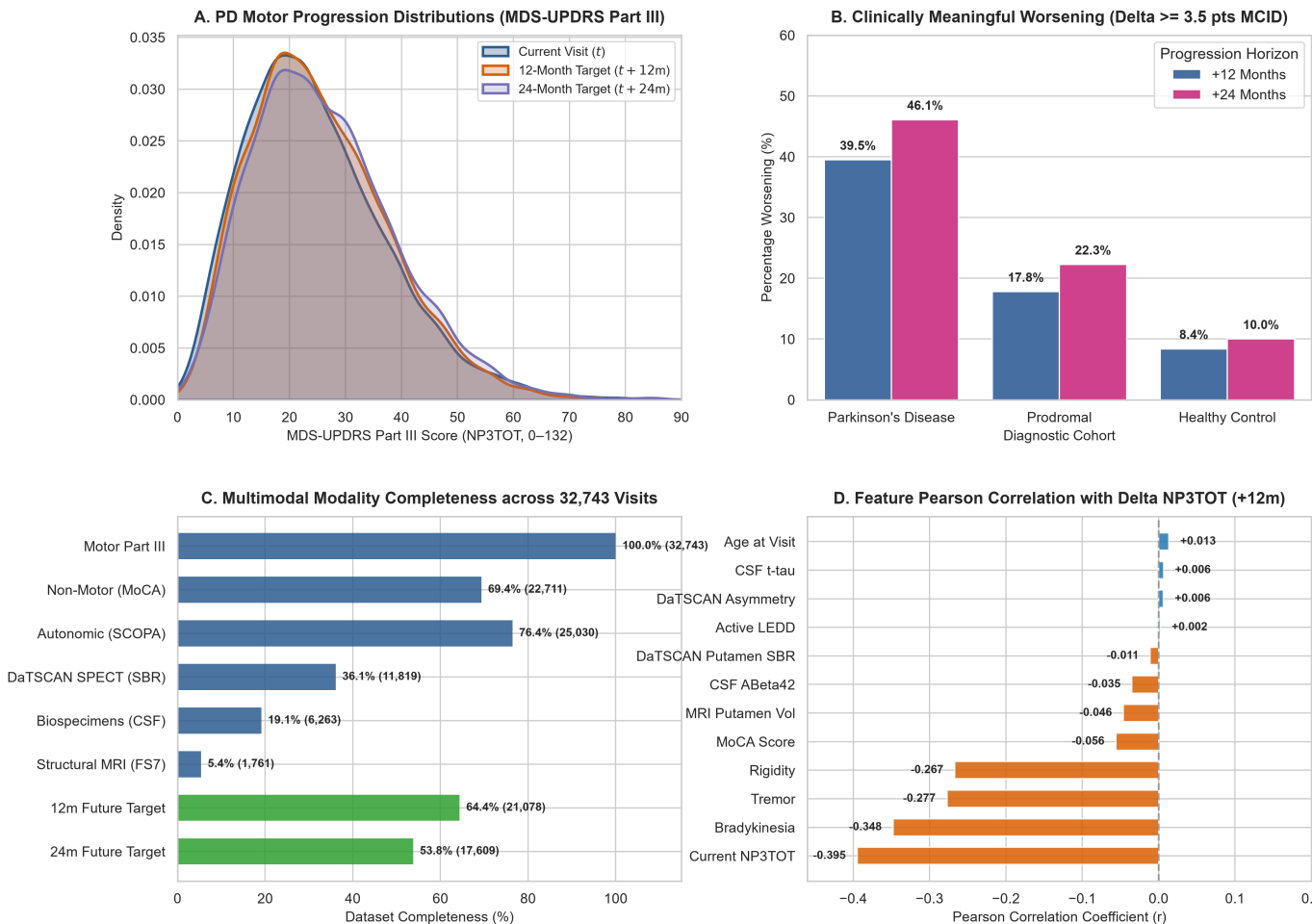


Figure 2: Multimodal Dataset Construction & Validation. (A) Probability density shift in NP3TOT over 12 and 24 months in PD. (B) Clinically meaningful worsening (MCID ≥ 3.5 points) across cohorts. (C) Data completeness across modalities. (D) Feature correlations with 12-month motor delta.

3. Technical Integrity, Parquet Artifacts & Phase 3 Roadmap

All pipeline outputs have been validated against our anti-leakage specification and serialized to high-performance Apache Parquet formats. Patient assignment was computed exactly once at baseline and strictly enforced across all tables.

Dataset Split	Total Rows	Unique Patients	Latest Tabular Visits	Primary Role in Capstone
Train Split	22,980 rows	3,787 patients	2,683 patients	Gradient boosting training, RNN/LSTM sequence fitting, representation learning.
Validation Split	4,831 rows	827 patients	574 patients	Hyperparameter tuning, early stopping, attention weight calibration.
Held-Out Test Split	4,932 rows	812 patients	583 patients	Strict blind evaluation (never seen during training or tuning).
Total Full Cohort	32,743 rows	5,426 patients	3,840 patients	Comprehensive multimodal longitudinal database (1.27 MB Parquet).

- Rigorous Anti-Leakage Architecture Enforced:**
 - Strict Patient-Level Partitioning:** No patient appears in more than one partition ($\text{Train} \cap \text{Val} = \emptyset$, $\text{Train} \cap \text{Test} = \emptyset$, $\text{Val} \cap \text{Test} = \emptyset$).
 - Temporal Arrow of Time:** No feature calculated at visit time t accesses observations from time $> t$. Future visits only supply the prediction targets (target_np3tot_12m, target_np3tot_24m).
 - Zero Split Contamination:** Scalers, imputers, and encoders are fit strictly on the training partition and transformed on validation and test.

Code Manifest & Local Git Commits

Commit Hash	Branch	Primary Artifacts Created & Committed
7161f9d	master	src/build_multimodal_dataset.py, src/visualize_data_engineering.py, data/processed/*.parquet (7 files), report/phase2_data_engineering_validation.png
f2e361f	master	src/cohort_edata.py, report/cohort_target_edata.png (Step 1 Exploratory Data Analysis)
a43d966	master	src/create_patient_split.py, data/processed/patient_split.csv (Phase 1 Canonical Split)

Immediate Next Steps: Phase 3 Modeling Execution

Phase 3, Step 1	Evaluation Harness (src/evaluate.py): Implement MAE, RMSE, Pearson r, R ² , ROC-AUC, PR-AUC, Sensitivity, Specificity, and Brier score, with medication subgroup stratification (ON vs. OFF).
Phase 3, Step 2	Model 1 Tabular XGBoost (src/train_model1_xgboost.py): Train static gradient boosting on latest visit features to establish the baseline and answer RQ1 (Single visit predictive power).
Phase 3, Step 3	Model 2 Sequential LSTM (src/train_model2_lstm.py): Train recurrent sequence network on multi-visit clinical history to test whether longitudinal trajectories beat static baselines.
Phase 3, Step 4	Model 3 Imaging Baseline (src/train_model3_imaging.py): Evaluate predictive signal from DaTSCAN SPECT striatal binding and FreeSurfer MRI subcortical volumes.